Computer Graphics

Colour & Illumination

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Colour Representation

- What do we write into the frame-buffer after identifying which pixels belong to a primitive?
  - Colour and intensity
- Colour represented using Red, Green, and Blue.
  - Visible portion of the electromagnetic spectrum:
    - From about 400nm (Violet) to 700nm (Red).
  - A bulb appears red if it emits light of λ in the red range
  - A flower appears orange because it absorbs light of all wavelengths except in the orange range.

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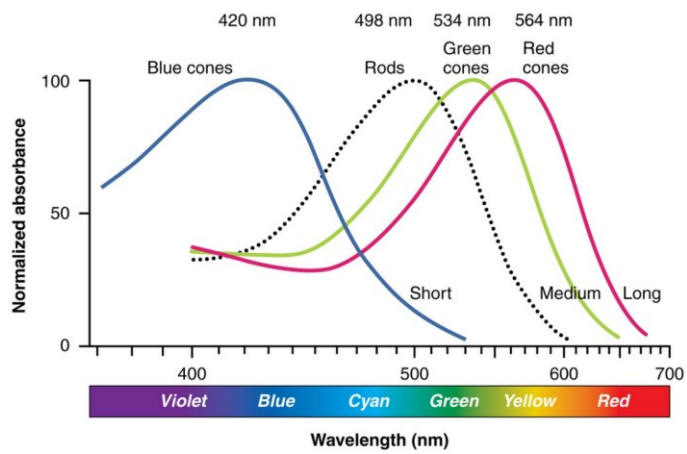
Human Color Perception

Photoreceptors

- The retina contains
  - 6 million cones (bright light/color)
  - 110 million rods (dim light/grayscale).
- **Rods** enable the "where" (movement, periphery, darkness).
- **Cones** enable
  - the "what" (color, detail, central vision).
  - Three types: Red, Green, and Blue.

Colour Sensitivity



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Tri-stimulus Theory

- All perceivable colours can be specified as weighted combination of primary colours R, G, and B.
- RGB define a set of basis vectors. Mix them in different measures to get any colour.
- Attractive, because a compact representation is possible.
- Problem reduces to: Find RGB values to approximate a given perceptual colour

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Colour Models

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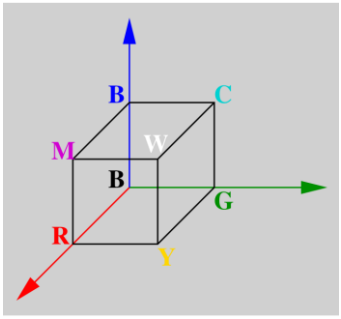
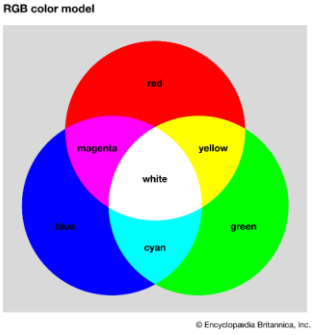
RGB Color Model

Additive Primaries

- Popular in graphics, due to CRT/LCD Monitors
- Values commonly normalized to range [0.0, 1.0].
- Externally 8-bits per colour channel
- Grays lie along the diagonal from (0, 0, 0) to (1, 1, 1).





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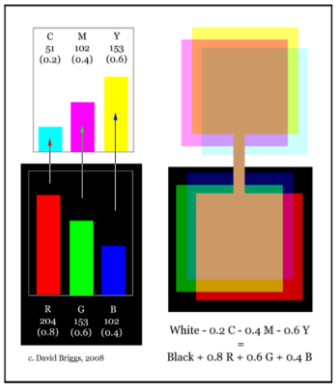
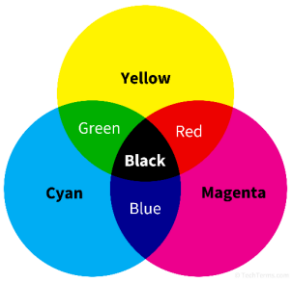
CMY(K) Color Model

Subtractive Primaries

- Subtract RGB from 1.0, you get CMY, respectively.
- Since black is common $\min(C, M, Y)$ is taken out as black or K component.
- Yields the CMYK colour model used in the 4-colour printing process.





RGBA Color Model

- Graphics systems (OpenGL, for example) use 4-component colours, with A being the α channel.
- Alpha measures the opacity of the colour. Equals 1 for completely opaque objects and 0 for totally transparent ones.
- A similar interpolation formula is used when an object with alpha is drawn on top of another object.
- If the Z-buffer test succeeds, the colours of the new polygon are blended into the frame buffer using α instead of replacing it entirely.

Illumination

An illumination model computes the lighting effects for a surface using the various optical properties that have been assigned to that surface

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Factors that influence Illumination

- **Light source properties**
  - Light sources emit light, and the model uses their **colour** and **directionality** (where the light comes from / how it is aimed).
  - Typical light types that determine directionality are directional, point, and spot lights.
- **Material (surface) properties**
  - Materials interact with light based on **reflectivity** and **colour** (the surface absorbs some wavelengths and reflects others).
  - Practically, illumination often splits reflectance into **diffuse** (dull/rough) and **specular** (shiny/highlight) behaviour.
- **Viewing geometry (camera)**
  - The appearance depends on the **viewing angle**:
- **Surface orientation (normals)**
  - For eg: Diffuse lighting depends on angle between surface normal and light direction), and correct effects require the **real normal** at the point.
- **Ambient and atmospheric effects**
  - Many models add **ambient light** (assumed present everywhere) to avoid completely black unlit regions.
  - They may also include **attenuation** (light diminishes with distance)

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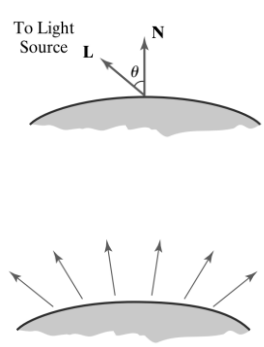
Material Properties

- **Colour**: the paint used on the surface of the material with which the object is made.
- **Reflectivity**: how the material interacts with light. Shiny, dull, grainy, etc.
- Hard to separate these effects in practice. Combined impact is observed.
- When light is incident on an opaque surface, part of it is reflected and part is absorbed. The amount of incident light reflected by the surface depends on the type of material.
- Types of materials:
  - Rough or grainy or Dull surfaces absorb more of the incident light. They scatter the reflected light in all directions referring to as **diffuse reflection**. Surface appears equally bright from any viewing angle.
  - Shiny materials reflect more of the incident light. some of the reflected light is concentrated into a highlight, or bright spot, called **specular reflection.**
  - Transparent surface, some of the incident light is also transmitted through the material.

Diffuse or Lambertian Reflection

- Objects (dull or rough) that obey Lambert's law of reflection: cloth, rough wall, etc.
- The normal component of light falling on it is absorbed by the object and is then reflected back equally in all directions.
- No preferred direction for reflecting light falling on it.
- Appearance of the point is independent of the view angle.
- Normal component of the light is proportional to $\cos\theta$.

Reflected Light $I \propto I_L \cos\theta$ in all viewing direction



Diffuse Illumination Equation

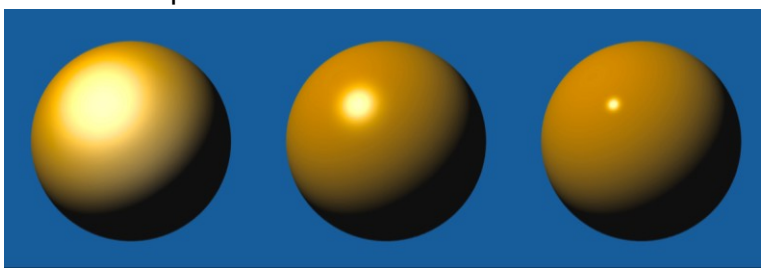
- Diffuse reflection at a point d, $I_d = k_d I_p \cos\theta = k_d I_p (\mathbf{N} \cdot \mathbf{L})$
  - I_p is the amount of incident light on the surface
  - k_d is the diffuse-reflection coefficient or the diffuse reflectivity. Higher for highly reflective surface [0.0 to 1.0]
  - θ should be between $0°$ and $90°$. Otherwise, the light has no effect. (Object is self-occluding!)
  - $\mathbf{N}$ is the normal at the surface point
  - $\mathbf{L}$ is the direction source direction vector
- For correct effects, the real normal at the point is necessary. (Normals are coming of use!)

Material Color

- What is the colour of an object?
  - The light it reflects. Rest is absorbed.
- A fully red object has colour $(1, 0, 0)$. It reflects all of the red falling on it and none of the green or blue.
- The spectral composition of the light falling on the object is needed.
- Illumination equation: $I_{d\lambda} = I_{p\lambda} k_d \mathrm{O}_{d\lambda} (\mathrm{N} \cdot \mathrm{L})$
  - $I_{p\lambda}$: incident light intensity/colour at that wavelength (light source contribution).
  - $\mathrm{O}_{\mathrm{d}\lambda}$: Object's diffuse colour/reflectance at that wavelength.
    - 0 represents complete absorption of light at that wavelength.
    - 1represents complete reflection of light at that wavelength.
  - $k_d \mathrm{O}_{\mathrm{d}\lambda}$: Diffuse colour of the object

Shiny Objects and Highlights

- Shiny objects (smooth metal, polished marble) behave differently.
- A highlight can be seen on them due to the light source. Highlight has the colour of the light source, irrespective of object colour
- Highlight moves with the viewing angle. Appearance of the object point depends on the view angle.
- This is called specular reflection.
- A mirror is an ideal specular reflector.



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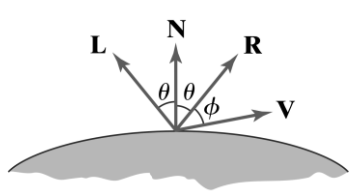
Shiny Objects and Highlights

- The bright spot, or specular reflection, that is seen on a shiny surface is the result of total, or near total, reflection of the incident light in a concentrated region around the **specular-reflection angle θ.**
- Objects other than ideal reflectors exhibit specular reflections over a finite range of viewing positions around vector **R**.
- Shiny surfaces have a narrow specular reflection range, and dull surfaces have a wider reflection range.

- **N**: Unit normal surface vector
- **R**: Unit vector in the direction of ideal specular reflection
- **L**: **U**nit vector directed toward the point light source
- **V**: Unit vector pointing to the viewer from the selected surface position
- *Φ:* the viewing angle relative to the specular-reflection direction **R**



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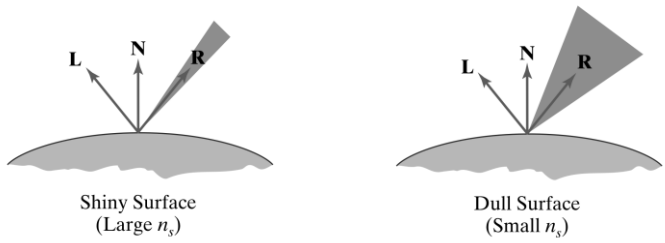
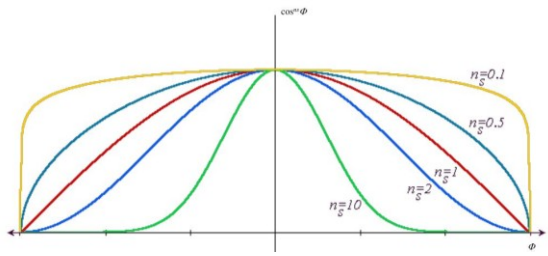
Phong Specular-reflection Model

- Sets the intensity of specular reflection proportional to $\cos^{n_s}\phi$
- n_s **Specular-reflection exponent** is determined by the type of surface.
  - A very shiny surface: Large n_s (say, 100 or more),
  - Duller surfaces: Smaller n_s (down to 1).
  - Perfect reflector: n_s is infinite.

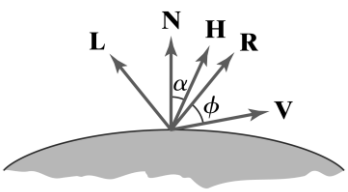




Specular Reflection Model

- Phong's model $I_s = k_s I_p cos^{n_s}\phi = k_s I_p (\mathbf{V} \cdot \mathbf{R})^{n_s}$
- ϕ is the angle between the reflection direction R and the view direction V.
- As n_s becomes larger, reflection becomes sharper.
- k_s is the specular reflection coefficient.
- Sometimes, specular colour $O_{\lambda s}$ is also given to the object!

Blinn-Phong Model

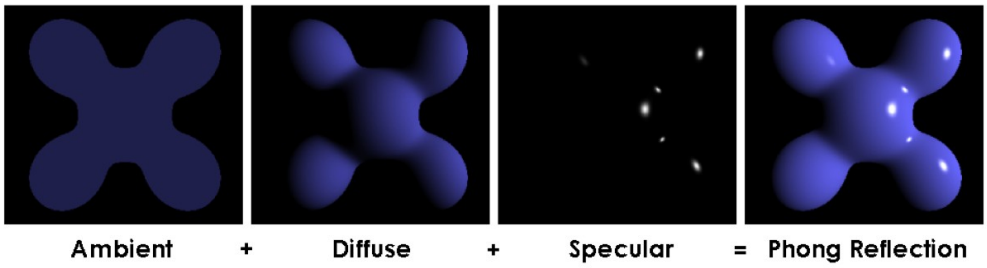


- **R** changes constantly: The reflection vector **R** depends on the surface normal **N**. Because **N** changes continuously across curved surfaces, **R** must be recalculated for every single vertex or fragment.
- Halfway vector, $\mathbf{H} = \frac{\mathrm{L+V}}{|\mathrm{L+V}|}$
- **H** can be constant: The halfway vector **H** is computed using only the light vector **L** and the view vector **V**. If both the light source and the viewer are distant, **L** and **V** are constant across the entire scene. This allows **H** to be computed exactly once per frame, yielding maximum computational savings.
- $I_s = k_s I_p cos^{n_s}\phi = k_s I_p(\mathbf{N}\cdot\mathbf{H})^{n_s}$

Ambient Light

- In graphics, some light is always present.
- This is called ambient light I_a, present everywhere.
- The net effect of all the light that reflects from the environment etc.
- This light helps see objects even when no explicit light source is present.
- Illumination equation: $I_\lambda = \mathrm{k_a} I_{\mathrm{a}\lambda} \mathrm{O}_{\mathrm{a}\lambda}$
  - k_a is the ambient reflection coefficient
  - $\mathrm{O}_{\mathrm{a}\lambda}$: Object's ambient reflectance at that wavelength

Combined Diffuse and Specular Reflections



Brad Smith

$$I_s = k_a I_a + k_d I_p(\mathbf{N} \cdot \mathbf{L}) + k_s I_p(\mathbf{V} \cdot \mathbf{R})^{n_s}$$

Atmospheric Attenuation

- The light that reaches the point $I = f_{att} I_p$
- Energy of a point light source falls off as the inverse square of d_L, the distance light travels from the source

$$f_{att} = \frac{1}{d_L^2}$$

- In practice
  - it doesn't work well as $1/d_L^2$ doesn't vary much if its far. And varies much if its closer.
  - Objects in real life are not illuminated by point source
- In graphics, $f_{att} = \min\left(\frac{1}{(c_1 + c_2 d_L + c_2 d_L^2)}, 1\right)$, c_1, c_2, c_3 are used defined constants
- Illumination equation: $I_\lambda = I_{a\lambda} k_a O_{a\lambda} + \boldsymbol{f_{att}} I_\lambda k_d O_{d\lambda}(N \cdot L)$

Multiple Light Sources

- We can place any number of light sources in a scene.
- For multiple point light sources, we compute the diffuse and specular reflections as a sum of the contributions from the various sources, as follows:

$$I = I_{ambient} + \sum_{l=1}^{n} I_{l,diffuse} + I_{l,specular}$$

$$I = k_a I_a + \sum_{l=1}^{n} I_l [k_d (\mathbf{N} \cdot \mathbf{L}) + k_s (\mathbf{V} \cdot \mathbf{R})^{n_s}]$$

Thank You